FaultLens

A diagnostic assistant for technicians repairing split air conditioners. It works through the job one check at a time, reads every kind of evidence a technician collects, stops unsafe steps, and says what to check next, or that it is not sure yet.

Open the live app: mrinal-datsci--faultlens-web.us-east.modal.run

The first visit can take up to a minute while the app starts. Model answers then take 10 to 30 seconds.

The problem

In an Indian summer a customer calls: the AC runs but the room will not cool. The technician must decide between a failed capacitor, a refrigerant leak, a stopped fan, a dirty coil or a blocked filter. A wrong guess wastes a part and a second visit; opening a live unit risks a shock. Most field technicians work alone, with a phone, and with no expert to ask.

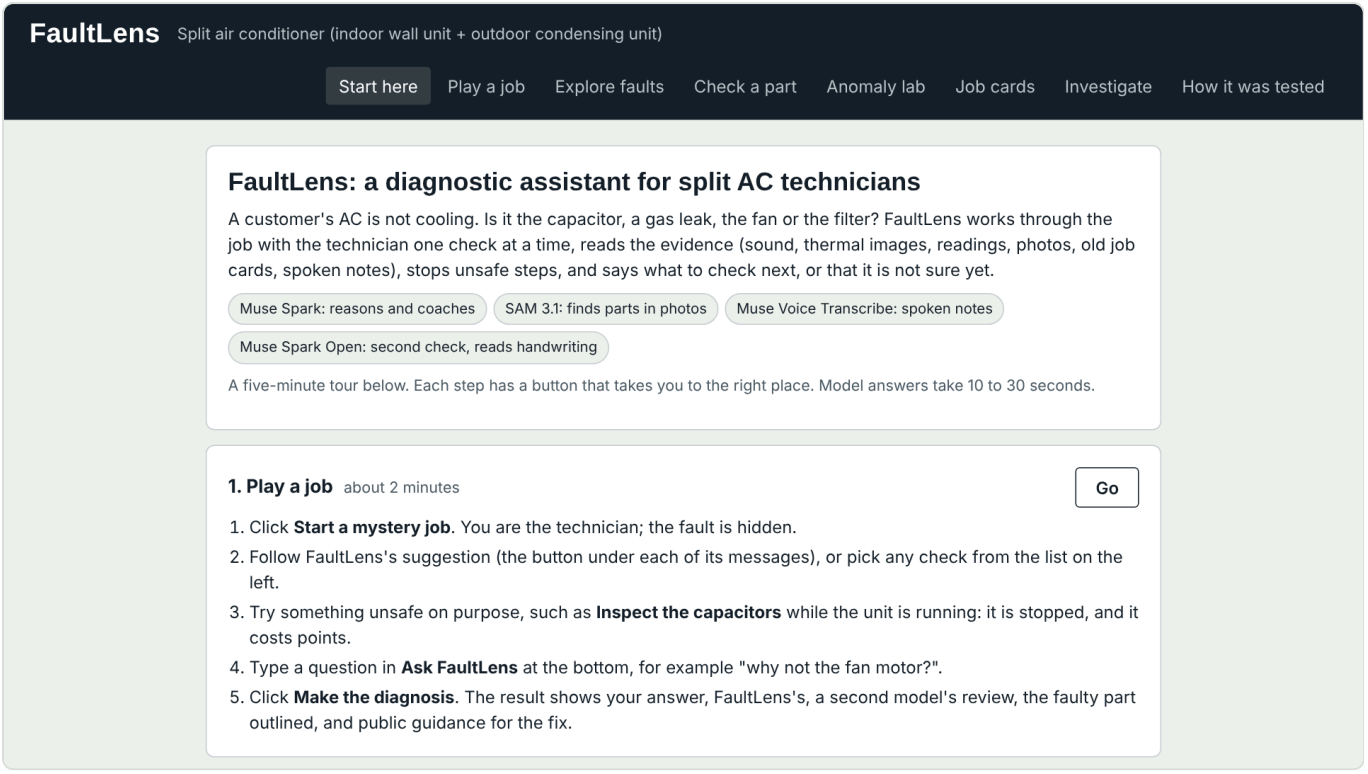
How Meta's models are used

MODEL	WHAT IT DOES IN FAULTLENS	WHERE TO SEE IT
Muse Spark	Reasons over all the evidence, coaches the technician step by step, checks parts in photos against healthy and damaged references, explains anomalies	Play a job, Check a part, Investigate, Anomaly lab
SAM 3.1	Finds and outlines parts in photos and on the exploded diagram of the unit	Investigate (with a photo), job results
Muse Voice Transcribe	Turns a spoken note into text, including Hindi and English mixed	Investigate, Record a voice note
Muse Spark Open	A second model reviews every diagnosis (on Baseten) and reads handwritten job cards (on Fireworks)	Job results ("Second check"), Job cards

Honest notes. Real faulty units could not be recorded, so the sound, thermal images and sensor logs are simulated from physical models of each fault and are labelled "Simulated" everywhere. Photos are generated or from the web and labelled with their source. On the public demo, model calls are limited per visitor and per day to protect the credits; if you reach the limit the page says so.

Getting around

The app opens on **Start here**: a short tour with a **Go** button for each step. The tabs along the top are the parts of FaultLens.

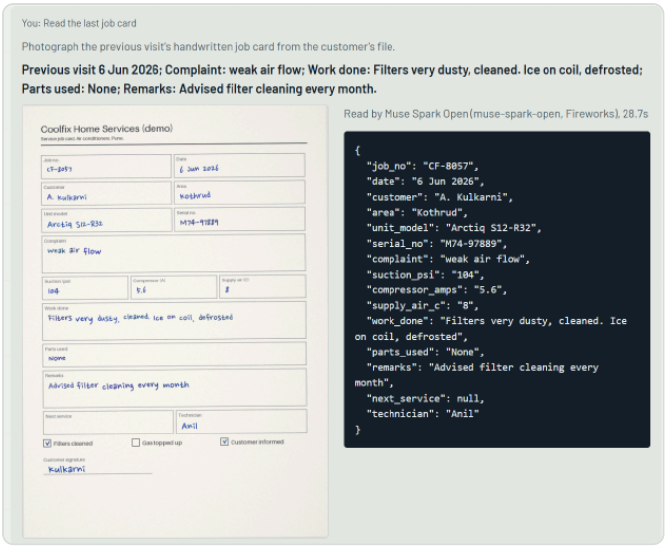


The Start here page, the first thing you see.

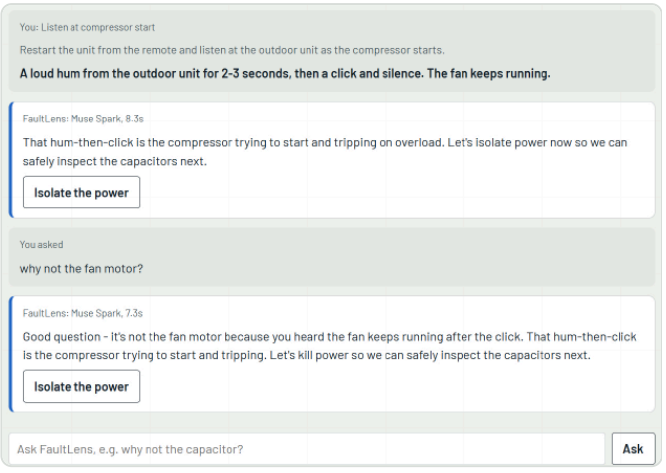
TAB	WHAT IT IS FOR	TIME
Play a job	Be the technician on a job with a hidden fault, coached by FaultLens	2 min
Check a part	Photograph a part: replace it or not, cause or symptom, wear and remaining life	1 min
Investigate	Upload what was recorded at the unit (sound, thermal, readings, photos, a voice note) and get ranked causes	1 min
Anomaly lab	Inject a fault into a healthy unit and watch FaultLens's own model find it	30 s
Job cards	Read a handwritten service card into fields	30 s
Explore faults	The full library: every fault's sound, thermal images, readings, alarm log and photos	optional
How it was tested	The measured results	1 min

1. Play a job about 2 minutes

1. Click **Start a mystery job**. You are the technician; the fault is hidden and the customer's complaint is shown.
2. Click **Read the last job card**: Muse Spark Open reads the previous visit's handwriting.
3. Follow FaultLens's suggestion (the button under each of its messages), or choose any check from the list on the left.
4. Try **Inspect the capacitors** while the unit is running. It is stopped, and it costs points.
5. Ask a question in **Ask FaultLens** at the bottom, for example "why not the fan motor?".
6. Click **Make the diagnosis**, choose a fault and click **Submit diagnosis**.



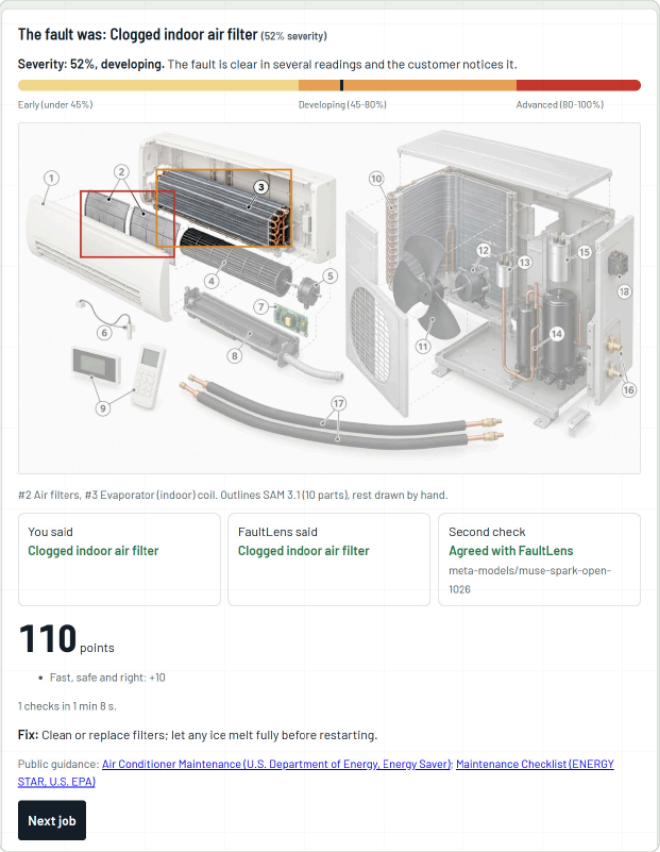
The last job card, read by Muse Spark Open: every field matched the card.



Push back and it answers from what you found: the fan kept running, so not the fan motor.

> **Look for:** it suggests one check at a time, refuses unsafe steps, and tells you when it is ready to conclude.

1. Play a job: the result



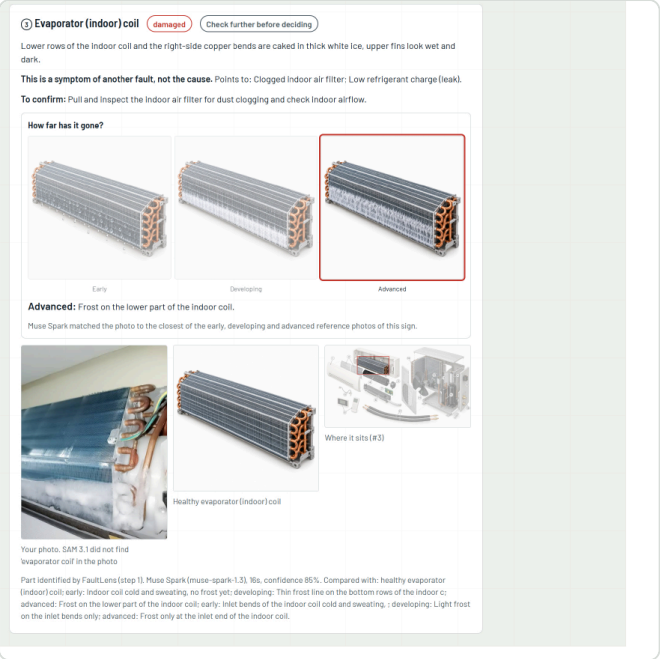
A finished job: your answer, FaultLens's, the second model's review, the faulty parts outlined, the fix and public guidance.

Look for:

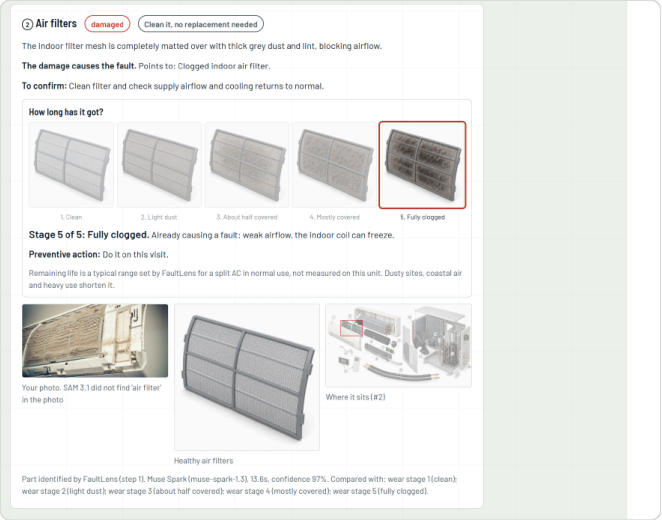
1. **Severity** on a scale from early to advanced; the photos and readings in the job match it.
2. **The diagram** outlines the cause (the filter) and the part that only shows the symptom (the iced coil).
3. **Second check:** Muse Spark Open, a different model, reviews FaultLens's call against the same evidence.
4. **The fix** links to public guidance (U.S. Department of Energy, ENERGY STAR, EPA, OSHA).
5. **Points** reward a fast, safe and right job; unsafe attempts and extra checks cost points.

2. Check a part about 1 minute

1. Pick a photo from the library on the left, or upload your own photo of an AC part (leave **Which part is it?** on **Not sure**).
2. FaultLens names the part, says whether it is damaged, what to do, and whether the damage **causes** the fault or is only a **symptom** of one elsewhere.
3. For filters, the outdoor coil, the fan hub and pipes it places the photo on a wear scale with the typical remaining life.



A real photo from the web, never seen before: an iced indoor coil. "This is a symptom of another fault, not the cause": check the filter or the refrigerant, do not replace the coil.

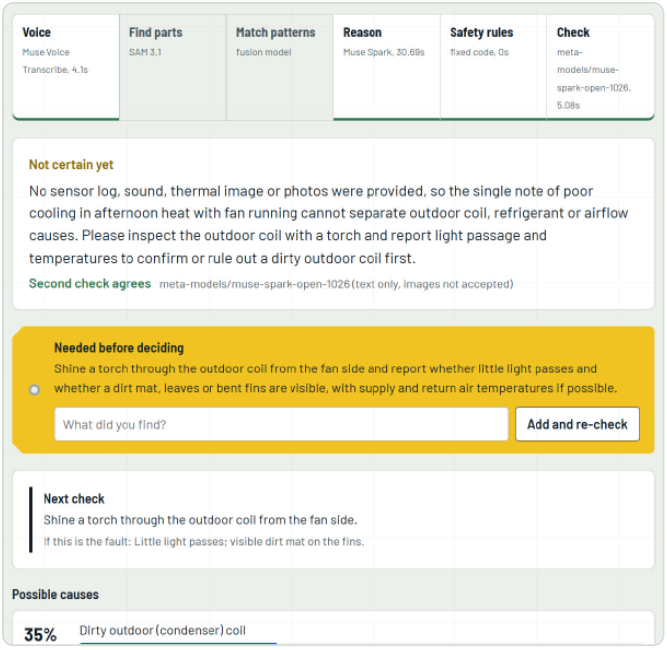


A real dusty filter: wear stage 5 of 5, already causing a fault, clean it on this visit. Preventive maintenance from one photo.

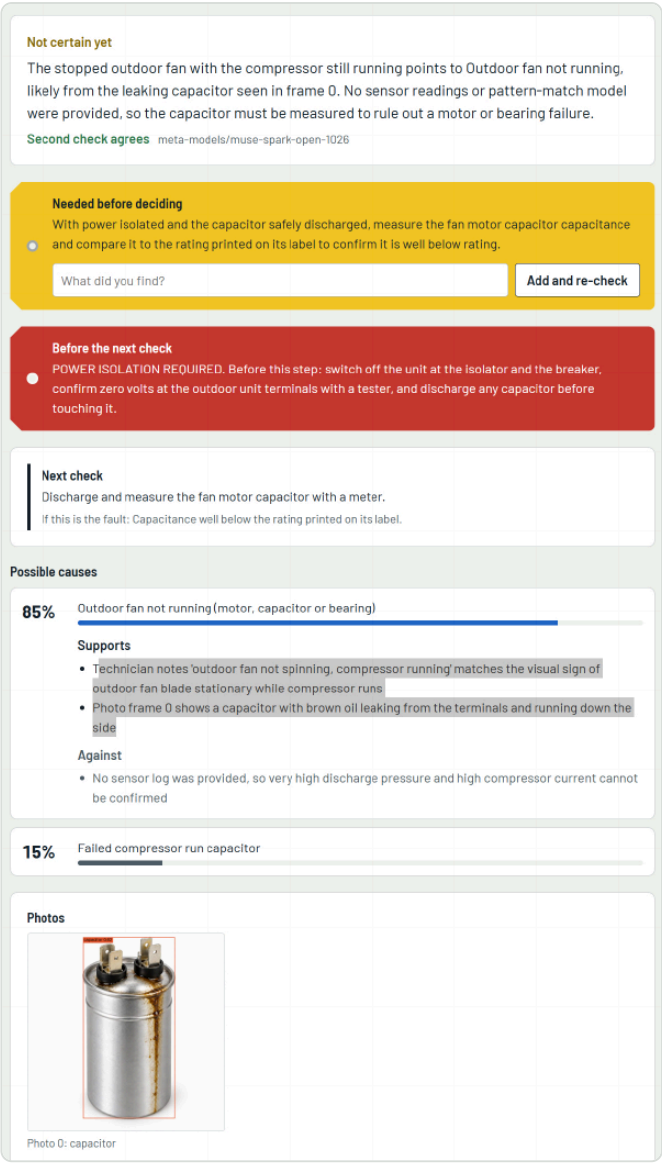
> **Look for:** on four real web photos (capacitor, filter, coil, fan) it named the right part every time, 85 to 98% sure.

3. Investigate about 1 minute

1. Choose a sample under **Or start from a library sample**, or upload files (sound, thermal images, a sensor log, photos), or click **Record a voice note**.
2. Click **Investigate**. The steps across the top light up as each model runs.



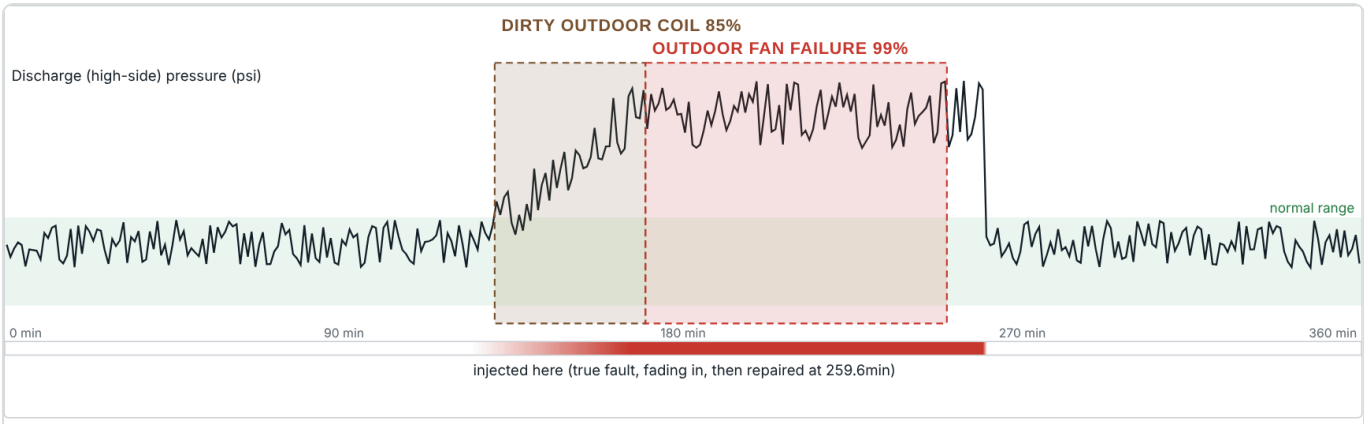
Only a spoken note: it does not guess. "Not certain yet", the causes ranked, and the one check that would settle it.



One photo: SAM 3.1 outlines the capacitor, Muse Spark sees the leaking oil, and a fixed safety rule adds "isolate the power" before the next check.

4. Anomaly lab 30 seconds

1. Pick a fault to inject. A healthy unit runs; the fault creeps in, stays, and is repaired.
2. FaultLens's own pattern model slides a window along the signal and labels every stretch. No model credits are used.
3. Click **Ask FaultLens to explain** for a plain-language summary from Muse Spark.



Injected: **Outdoor fan not running (motor, capacitor or bearing)** from 124.3 to 259.6 min. Model called: **Dirty outdoor coil** (85%, 130-170 min), **Outdoor fan failure** (99%, 170-250 min).
right fault found, 59% of the true stretch covered, first caught 45.7 min after it began · named the wrong fault for 40 min inside the stretch (Dirty outdoor coil), usually while it was still mild · **no false alarms**.

Ask FaultLens to explain Muse Spark reads this run and explains it. One model call.

Simulated signal, from the same physical models the pattern model was trained on. New unit and new fault timing every run.

An outdoor fan failure injected into 6 hours of readings. While the fault is mild the model first calls it a dirty outdoor coil (both raise the head pressure), then names it correctly.

5. Job cards 30 seconds

Pick one of the 14 handwritten service cards (or upload a photo of one). Muse Spark Open reads the handwriting into fields, so a unit's history is not lost. Measured on 4 cards: 13 to 15 of 15 fields right each time; the misses were model and serial codes (1 read as I).

6. How it was tested

Every number here was measured with live models and can be re-run from the code. The **How it was tested** tab shows the same results.

Full investigation, 14 cases (one easy and one hard per fault)

METHOD	CORRECT	HARD CASES	CORRECT WHEN CONFIDENT
Sensor rules only (baseline)	50%	29%	42%
FaultLens, sensors only	79%	57%	80%
FaultLens, sensors + note + alarm history + second check	**93%**	86%	**100%**

In the hard cases the fault is mild and one key sensor is missing. The one miss was a mild refrigerant leak called healthy, and FaultLens said it was not sure. Every time it said it was confident, it was right. 99% of the evidence it quoted was checked automatically and found in what it was given.

Other measurements

WHAT	RESULT
Check a part, 15 library photos (each left out of its own references)	Right part 15 of 15 · right condition 13 of 15 · wear stage within one 4 of 4 · severity level exact 7 of 9
Check a part, 4 real web photos	Right part and right call 4 of 4
Pattern model, 420 unseen simulated units	98% with all four signals combined, 94% on mild faults; no single signal above 90%
Job cards, handwriting	13 to 15 of 15 fields per card
Anomaly lab, 20 runs per fault	Sensors: right fault in 20 of 20 for every fault; sound: 20 of 20 for five faults, 10 of 20 for the clogged filter